Vinith Ayapilla

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SUMMARY

Engineer and evaluation researcher moving into AI safety. I'm drafting a proposal for my own project on how LLM safety behavior shifts across languages, and I've run safety benchmarks on open-weight models (an abliterated model failed every Vending-Bench trial). At the TIMSS & PIRLS International Study Center, I research whether LLM judgments are reliable and valid, and build LLM pipelines on self-hosted open models. Before that, I spent two and a half years at ZS Associates on pharma commercial data.

SKILLS

Evaluation & measurement: LLM-as-judge, comparative judgment, Bradley-Terry, psychometrics/IRT, safety benchmarks, evaluation harnesses, human-in-the-loop review
LLM systems: RAG, agents, coding agents, on-prem open-model deployment, quantization, fine-tuning, model routing, OpenAI & Gemini APIs, Claude code
Front-end: TypeScript, React, JavaScript, HTML/CSS, data visualization, WCAG 2.2 AA
Data: Python, SQL, R, Redshift, Airflow, statistical reporting, Power BI, Tableau
Business: Strategy, go-to-market, commercial pharma data, stakeholder management

EXPERIENCE

IEA TIMSS & PIRLS *Chestnut Hill, MA*

Technical Dev/Consultant - AI Evaluation & Measurement June 2026 – Current

- Rebuilt the interactive international results site on React/TypeScript, including the report landing page, accessibility, and phone layouts (shipped Oct 2026), taking automated accessibility errors on audited pages from 110 to 0 (WCAG 2.2 AA)
- Lead the research program on LLM comparative judgment for item-difficulty estimation, with two paper submissions (see Research)
- Building a three-decade TIMSS item dataset (hundreds of thousands of rows) to post-train an open-source model to predict item IRT parameters (in progress)
- Translate cross-national assessment results into recommendations for education ministries.

AI Engineer Sep 2025 – June 2026

- Digitized decades of print-only assessment items with LLM-assisted transcription and standardized them into one schema across cycles
- Built an evaluation harness scoring 10,000+ open-ended student responses against ground truth, with a human-in-the-loop layer catching data-quality and sourcing errors
- Led the move from proprietary OpenAI APIs to self-hosted open models for EU GDPR compliance, with routing across quantized open-weight models and API fallback, cutting per-question cost from ~$60 to near-zero

Boston College *Chestnut Hill, MA*

Sr. Data Analyst | Awards/Grants Intake Team | Graduate Assistant Nov 2024 – Sep 2025

- Ran the data side of research grant intake, tracking budgets and award data in Oracle PeopleSoft, as primary data advisor to the Vice-Provost's office
- Built automated reporting and Power BI dashboards that cut grant reporting time by 75%

ZS Associates *NCR, India*

Associate | Business Technology Jan 2022 – Jun 2024

- Migrated the data science team's experimental data from a legacy system to a cloud platform (CDR-SCE) using SAS and Shell
- Built a real-time geo-analytics tool for territory planning, driving $500K in new annual revenue
- Redesigned Key Account Manager workflows, cutting administrative overhead by 90%
- Diagnosed root-cause flaws in a failing client data pipeline and rebuilt it as a proof of concept, cutting manual QC by 50%; orchestrated pipelines in Airflow and Azkaban with severity-based alerting
- Ran commercial data strategy for 3 specialty drug launches across Sales, Marketing, and IT, for assets representing 80% of client revenue

PROJECTS & RESEARCH

Cross-Lingual Safety Gaps *In progress, proposal drafted*

Mapping how and why LLM safety behavior (refusals, harmful outputs, over-refusals) shifts across languages, comparing machine-translated, human-translated, and natively written prompts across US and Chinese frontier models to separate translation effects from language and model-origin effects

AI Safety Evaluation - Uncensored vs. Baseline Model Benchmarking *August 2026*

Ran a local reimplementation of Vending-Bench Mini alongside AgentHarm, StrongREJECT and WMDP to compare an uncensored open-weight model (Gemma-4 26B) against its stock baseline, using Gemini as judge under hardware-constraints.

The abliterated model performed worse - went out of business in all trials compared to the original which went out of business only about ~50% of the time and this was expected due to manual weight surgery on the refusal vector.

Auditing LLM Comparative Judges Against an External Latent Criterion *NeurIPS 2026(not accepted, being revised)*
Audited an LLM pairwise judge across 12,804 item pairs presented in both orders against withheld operational IRT difficulty. Found 11.06% order-instability concentrated among near-ties, accounting for ~95% of observed intransitivity; showed that filtering unstable judgments produced near-perfect transitivity without improving criterion validity - a more internally consistent judge is not a more valid one. Reviewers flagged the scale as the reason for non-acceptance.

Designing Efficient LLM Comparative Judgment Networks for Item Difficulty Estimation *NCME 2027 submission*
Evaluated how anchor information and comparison-network design affect LLM-based item-difficulty estimation across 1,351 Bradley-Terry fits over 28 network designs. Showed that explicit calibrated anchor information improves scale recovery without improving rank recovery, and that sparse networks (22% of the full comparison budget) retain 98% of rank correlation but produce badly overdispersed estimates, i.e. efficient ranking designs shouldn't be mistaken for calibrated measurement.

Kosovo Education Policy Analysis: The Enthusiasm–Achievement Paradox *July 2026*
Authored an independent multi-pillar policy analysis cross-referencing IEA PIRLS/TIMSS achievement data with World Bank and Kosovo labor-market data to explain the gap between high student engagement and low cognitive-skill conversion which spans across socioeconomic, schooling infrastructure, curriculum, and brain-drain dimensions and proposes a policy roadmap.

EDUCATION

Boston College | Carroll Graduate School of Management *Boston, May 2026*
MBA | STEM Track | Strategy | Data | GMAT 740 | Recipient of Dean's Scholarship | CGPA - 3.64/4.00

Shiv Nadar University *India, May 2022*
Bachelor of Technology in Electrical & Electronics Engineering | Minor in Big Data Analytics